

CI Assignment 2

Coded apertures



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1 Introduction

In this lab, we explore the use of coded apertures to mitigate the loss of image detail that occurs when objects are not perfectly in focus or when motion blur is present. Traditionally, a circular aperture may introduce zeros in the Fourier domain making it impossible to recover all image details through deconvolution. By contrast, coded apertures distribute energy more evenly across frequencies, helping preserve high-frequency information and leading to sharper reconstructions.

2 Blur simulation and recover of focused image

In this first section we have to artificially blur the image with an specific aperture (in this case is circular but in section 4 we will change it) to then recover the original image after that. First we need to understand the code and identify the different parts of this first part.

- $f_1 = \text{zDefocused}(I, \text{kernel}, \sigma, \text{flag});^1$. This is the part where the original image f_0 gets blurred by applying a kernel k (that consists in re-scaling the aperture interpolating with the nearest neighbor without introducing new values). It behaves different depending on the **flag** variable that can be -1 or 1.
 - if **flag** is set to 0 it uses spatial domain filtering.
 - if it is set to 1 computes a circular convolution using fast fourier transforms.

In both cases it uses gaussian noise, appearing clear boundaries because of it. (Figure 1).



Figure 1: On the left, blurred image with flag set to 0. On the right, same blurred image but with flag set to 1. Both of them using $\sigma = 0.05$ and 10 as blur size.

- $\hat{f}_o = \text{zDeconvWNR}(f_1, \text{kernel}, c)$ This is how we recover the original image by applying Wiener deconvolution (Eq. 1).

$$I = \frac{|\mathcal{F}(k)|^2}{|\mathcal{F}(k)|^2 + \frac{1}{\text{SNR}(w)}} \cdot \frac{\mathcal{F}(B)}{\mathcal{F}(k)} \quad (1)$$

¹We can interpret this as a spatial filter where each pixel is a weighted combination.

where:

- $\mathcal{F}$ denotes the Fourier transform.
- $B = f$ is the observed (blurred) image.
- k is the known blur kernel (PSF).
- $SNR(w)$ is the signal-to-noise ratio in the frequency domain.

The parameter C in **zDeconvWNR** acts as a regularization term and is related to the inverse SNR:

$$C = \frac{\sigma^2}{A} = \frac{1}{SNR(w)} \quad (2)$$

This regularization prevents amplification of noise in frequencies where the PSF has low energy.

Then we can try different parameters to evaluate the results such as the amount of gaussian noise in the image (σ) and the blur size (Fig. 5).



Figure 2: $\sigma = 0.0001$ and 10 blur size

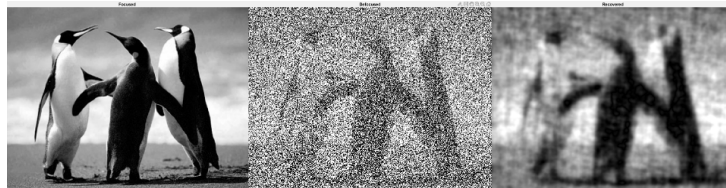


Figure 3: $\sigma = 2$ and 2 blur size

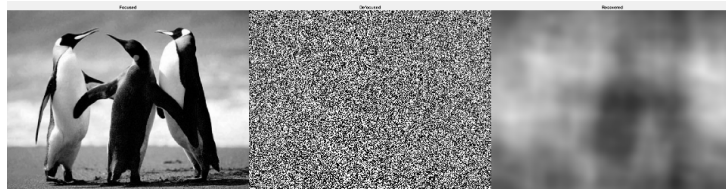


Figure 4: $\sigma = 10$ and 10 blur size

Figure 5: Comparison between both extreme cases where we can put or high noise (middle) or blur size (upper) and we can put both high (bottom)

What we can notice on the previous figures is that we can recover easily the original image if we have high noise rather than if we have high blur size. This is because the gaussian noise adds random

variations to the pixels and most of the algorithms that we are using suppress it effectively. In contrast, blurring spreads pixel values over a region and that leads to loss information. This means that, unlike noise that can be filtered, blur removes high frequency details that cannot be reconstructed.

Additionally, for a more detailed comparison beyond visual inspection, we evaluated the quality of the recovered images using objective metrics. Specifically, we compute the Peak Signal-to-Noise Ratio (PSNR) [1] and the Structural Similarity Index Measurement (SSIM) [2] which quantify the difference between the original and recovered images normalized into value range 0 and 1 taking into account contrast and luminance.

- The PSNR measures the ratio between the maximum possible pixel intensity and the error introduced by degradation and restoration. A higher PSNR indicates better reconstruction quality (3).

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (3)$$

- MAX_I is the maximum pixel value in the image (255 for 8-bit images).
- MSE is the Mean Squared Error, which measures the average difference between the original and reconstructed image (4).

$$MSE = \frac{1}{m \times n} \sum_{i=1}^m \sum_{j=1}^n (I(i, j) - K(i, j))^2 \quad (4)$$

where $I(i, j)$ is the pixel of the original image, and $K(i, j)$ is the pixel of the recovered image.

- SSIM, unlike PSNR, considers structural information in the image, such as luminance, contrast, and texture, and provides a more perceptually information of the quality of the image. It is a value between -1 and 1, where 1 indicates perfect similarity between the two images, and values closer to -1 indicate greater dissimilarity (5).

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (5)$$

By analyzing these metrics across different noise levels (σ) and blur sizes, we can identify how each parameter affects the restoration process and confirm whether high blur introduces more significant degradation than high noise. So, finally, we implemented a program that iterates over a set of 20 values of noise levels and blur sizes that obtain its PSNR and SSIM to then show a mesh with all values (Figure 10).

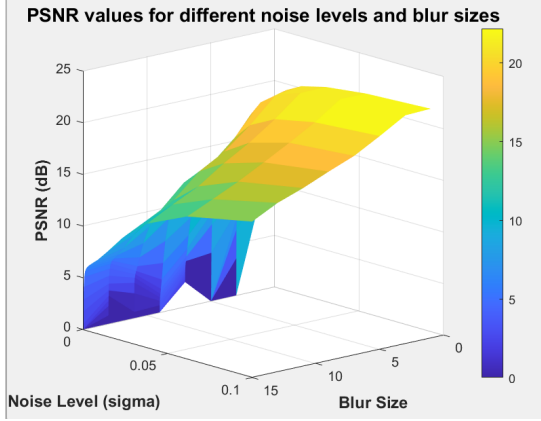


Figure 6: PSNR

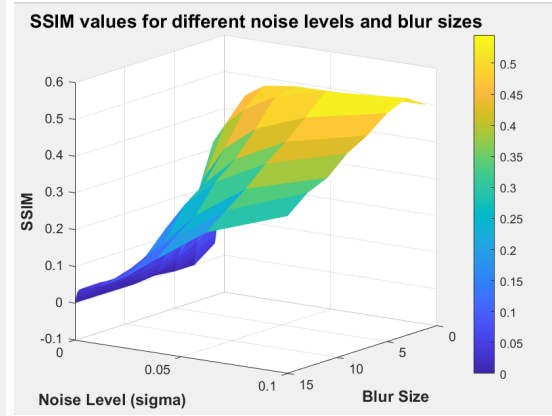


Figure 7: SSIM

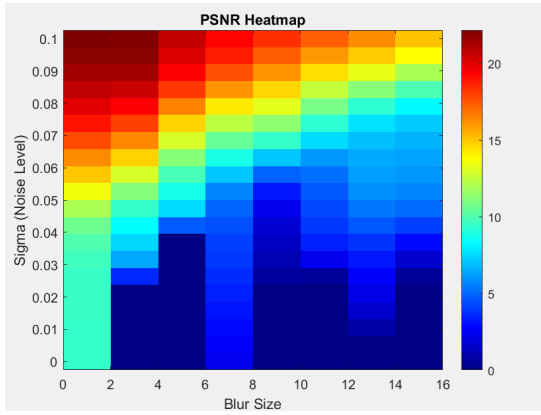


Figure 8: Heatmap for PSNR

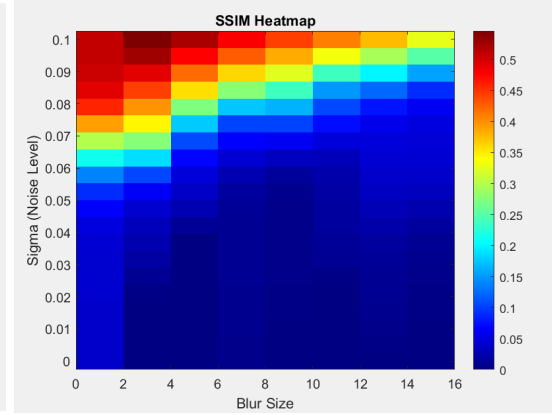


Figure 9: Heatmap for SSIM

Figure 10: Metrics grids to evaluate different parameters of noise and blur size

Those graphs represent what is the PSNR and SSIM values of the comparison between our reference image (original) and the recovered one with some σ and blur size values (each tile).

We can notice on those graphs show how restoration quality (measured by PSNR and SSIM) varies with changing noise level (sigma) and blur size in the image. It is observed that for low noise levels and blur sizes, both PSNR and SSIM are higher (better quality), while with increasing noise or blur, the quality metrics decrease. This indicates that the Wiener deconvolution method with the prior matrix performs best under low degradation conditions, and becomes less effective as the image degrades further due to increased blur and noise.

With this conclusion we can proceed to the next part which is to compare different deconvolution algorithms, specifically the Lucy-Richardson deconvolution and Wiener deconvolution without priors. Lucy-Richardson deconvolution is an iterative method that starts from a statistical model and refines the estimate of the original image at each iteration, based on the probability that each pixel comes from the sharp image. Wiener deconvolution without priors applies the classical Wiener filter, which assumes

a power spectrum of the noise and the image (without specific prior information) and seeks to minimize the mean square error in the Fourier transform domain.

We can see some results of the penguins image applying the same parameters to the three deconvolutions in the figure 14.



Figure 11: Wiener deconvolution without priors

Figure 12: Wiener deconvolution with priors

Figure 13: Lucy-Richardson with 10 iterations

Figure 14: Metrics grids to evaluate different convolutions algorithms under $\sigma = 0.01$ and 6 blur size

We can notice that with low noise levels and a blur size that is not too small, the Wiener methods are less effective, as mentioned in the previous paragraph. In contrast, the Lucy-Richardson method appears to be more robust when the image is degraded. However, it is important to note that using too many iterations in the Lucy-Richardson method can introduce artificial noise that was not originally present.



Figure 15: Wiener deconvolution without priors

Figure 16: Wiener deconvolution with priors

Figure 17: Lucy-Richardson with 10 iterations

Figure 18: Metrics grids to evaluate different convolutions algorithms under the best case for the Wiener deconvolution. In this case we can see that Wiener deconvolution with priors alongside Lucy deconvolution are the best options

Lucy is not perfect as it is very sensible to the noise level so, as we already know how Wiener deconvolution behaves with different aparameters. We have done the same experiment with Lucy deconvolution with 10 iterations to see the differences when changing the blur and sigma parameters (Fig. 23).

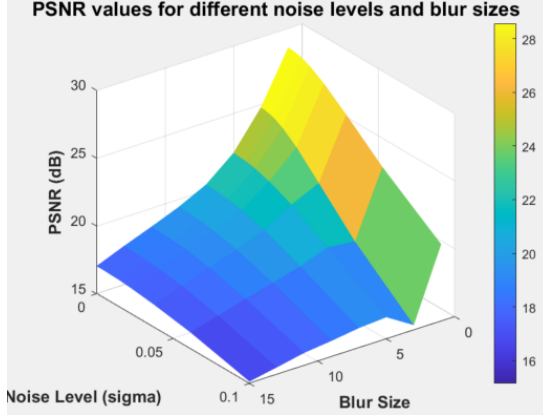


Figure 19: PSNR

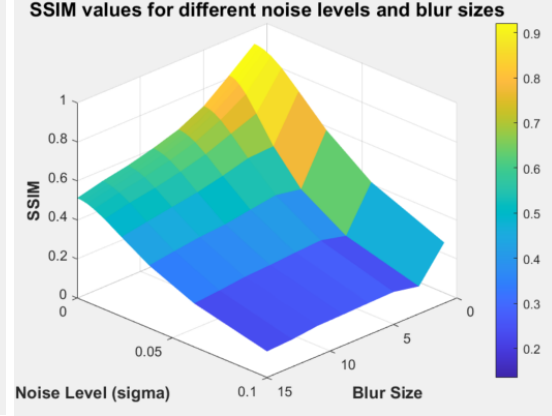


Figure 20: SSIM

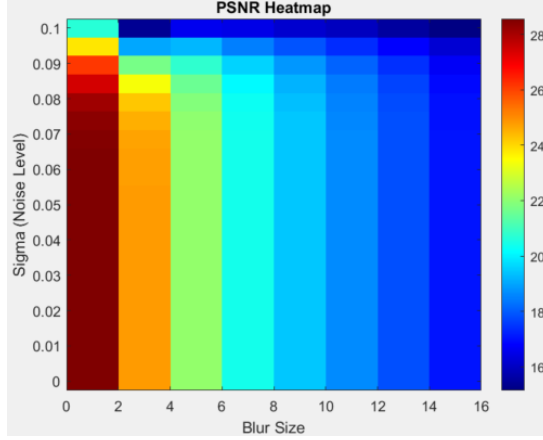


Figure 21: Heatmap for PSNR

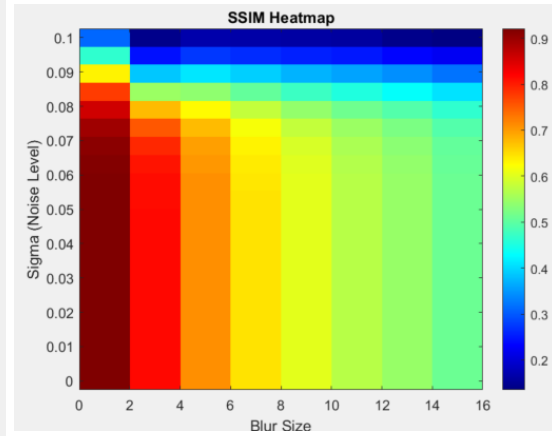


Figure 22: Heatmap for SSIM

Figure 23: Metrics grids to evaluate different parameters of noise and blur size with Lucy-Richardson deconvolution

From the heatmaps, we can observe that the deconvolution process becomes more sensitive as noise levels rise, while it remains notably robust across various blur sizes when the noise level is fixed.

In general, we can observe that the prior information significantly influences the recovery of the original image. For instance, if we increase the blur size without raising the noise level, the PSNR values tend to drop, as seen in the heatmaps. This behavior is logical because, while the Lucy deconvolution method is more robust against increasing blur compared to the Wiener method, it eventually starts to fail when the blur becomes too extensive (Figure 28).



Figure 24: 2 blur size, practically the ideal case



Figure 25: 5 blur size

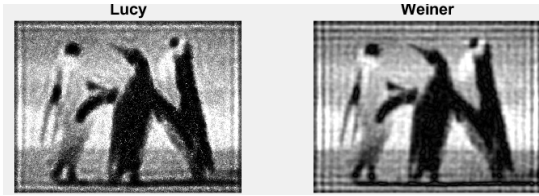


Figure 26: 10 blur size

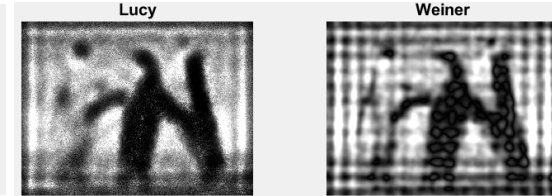


Figure 27: 20 blur size

Figure 28: Weiner versus Lucy-Richardson comparison with $\sigma = 0.1$ and different blur sizes

Demonstrating with the previous figure that, for the ideal case, Weiner can be a good option.

3 Power spectra

In the Fourier domain, the power spectrum represents the distribution of signal power across different frequency components. Mathematically, if $F(k)$ is the Fourier transform of a function $f(x)$, then the power spectrum is given by:

$$P(k) = |F(k)|^2$$

where:

- $|F(k)|$ is the magnitude of the Fourier transform at frequency k , - Squaring the magnitude gives the power associated with that frequency.

The power spectrum essentially tells us how much energy each frequency contributes to the overall signal.

The power spectrum is a crucial tool in signal processing and inverse problems because it helps analyze and reconstruct signals where zeros in the Fourier domain cause information loss.

Using circular aperture, we can check that we have significant information near zero (29), which means we were able to recover less information. As we can tell from 30 the information is lost after applying a blurred effect and we can't recover all information on the zeros of the Fourier domain. This will be improved by using different types of apertures as will be shown in the next section.

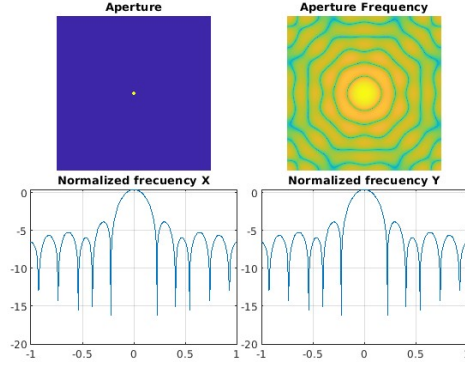


Figure 29: Circular aperture using Lucy deconvolution under a $\sigma = 0.001$ and a blur size of 5.

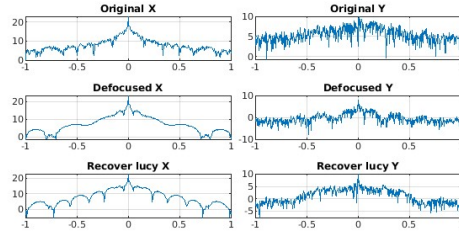


Figure 30: Fourier domain of image after applying blur with $\sigma = 5$ and after Lucy-Richardson deconvolution apply power spectrum.

4 Coded apertures

Coded apertures are designed to recover focused images while minimizing the loss of information during the convolution and deconvolution processes. Unlike traditional circular apertures, which can result in the loss of high-frequency details, coded apertures encode additional spatial information. This can be clearly observed by comparing the results in the Figure 35 [?].



Figure 31: Levin

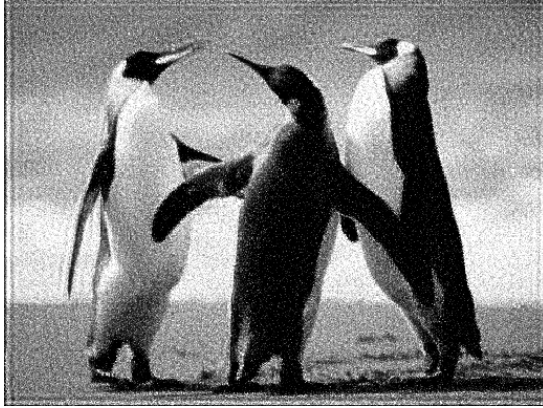


Figure 32: Raskar

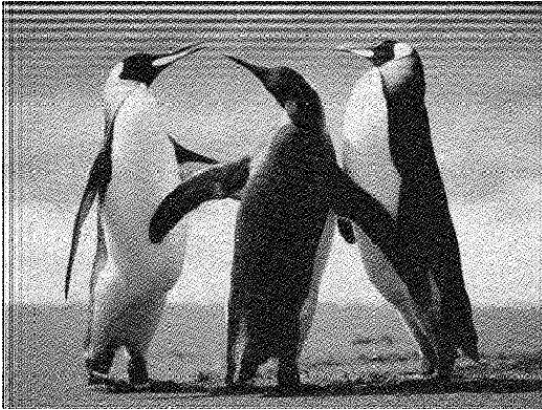


Figure 33: Zhou



Figure 34: Circular

Figure 35: Different apertures using Lucy-Richardson deconvolution under a $\sigma = 0.01$ and a blur size of 5

As we already know, the coded apertures have a pattern on that blocks part of the incoming light. Even if we have everything in focus we have some imperfections (looking at the PSF of the circular aperture as shown in the figures of the previous section).

As mentioned above, recovering the original frequency components is challenging because certain frequencies are completely removed when an image is blurred. In the case of circular apertures, some frequency components in the Fourier domain are multiplied by zero, leading to a total loss of information at those frequencies. The objective is to recover as many of the original frequency components as possible.

To solve this problem, coded apertures are used. Unlike circular apertures, coded apertures alter the shape of the point spread function (PSF), which is the system's response resulting in fewer frequency components being reduced to zero or near-zero values in the Fourier domain. This change allows us to recover more information compared with regular apertures, even though some attenuation still occurs (Figure 39).

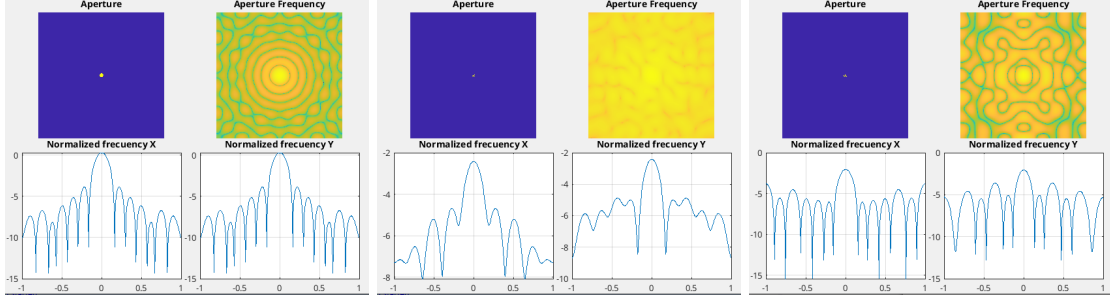


Figure 36: Regular aperture

Figure 37: Raskar aperture

Figure 38: Levin aperture

Figure 39: Normalized frequencies in the Fourier domain.

From these images, we can see how each aperture's frequency response differs in the Fourier domain. The regular (circular) aperture exhibits zeros or very low amplitudes at certain frequencies (in general), that may cause some loss of information there. In contrast, the coded apertures (Raskar, Levin) have more uniform frequency responses and less closer to 0, which is the main problem. This means they retain more information across the spectrum.

Also this frequencies depends on how the blur affects the image. The more blurred we have the image, the less high frequencies we will have so that will be more complicated to recover (Figure 44).

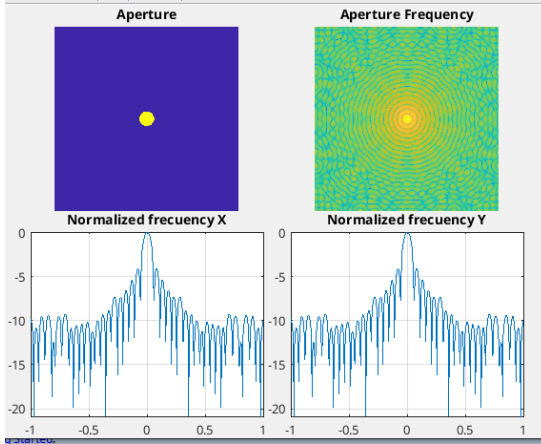


Figure 40: Fourier domain using circular apertures.

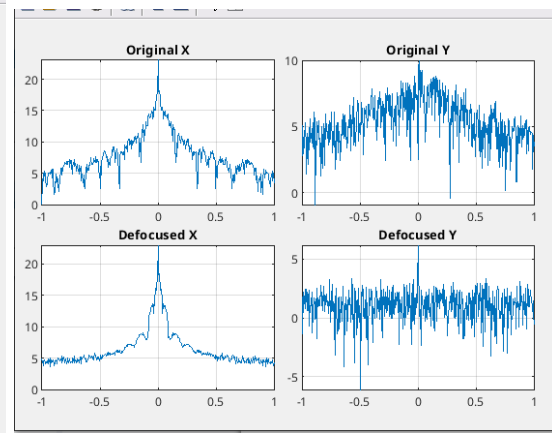


Figure 41: High blurred image frequencies using circular aperture.

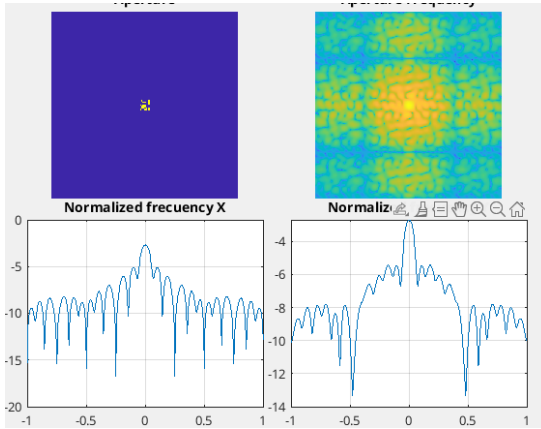


Figure 42: Fourier domain using raskar apertures.

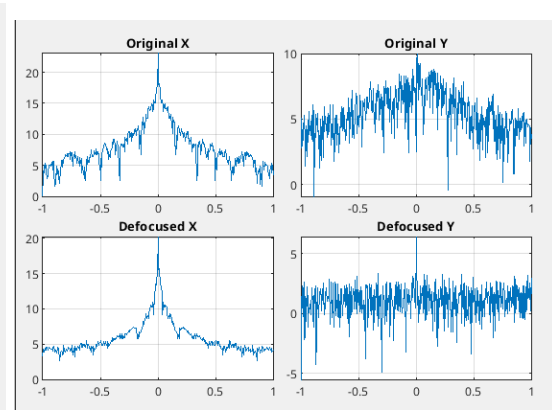


Figure 43: High blurred image frequencies using raskar aperture.

Figure 44: Power spectrum of the original and recovered image using circular and raskar aperture and high blur size

Notice on this previous figure how in the original images the original graphs shows the frequencies that the original image had and the defocused graph shows the same power distribution for the same image but blurred. Typically we will see that high-frequency components (those far from the center in the frequency domain) are highly reduced because blurring reduces the energy in the higher-frequency range (commented in previous sections). In the circular aperture the Fourier domain shows that noticeable drops in some frequencies indicating that some components may have lost or are heavily attenuated. In contrast, looking at Raskar's aperture Fourier domain suggest a more uniform spread of frequency information, which indicates a better preservation of detail. If we take a look at the original images we will see this information visually in the figure 47.



Figure 45: Image recovered using circular aperture.

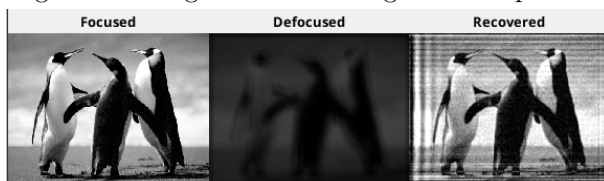


Figure 46: Image recovered using coded aperture (Raskar).

Figure 47: Final result of applying a coded aperture versus applying a regular aperture. Notice how the coded aperture is able to recover most of the information and the regular aperture introduces artifacts on the recovered image.

However, coded apertures introduce a trade-off: they allow less incoming light, which increases noise due to a lower signal-to-noise ratio. This creates an optimization challenge where the goal is to maximize the incoming light while minimizing the loss of frequency information in the Fourier domain.

5 Color image

In the first section of this report, we discussed the recovery results of various images with different blur sizes and noise levels using single-channel (grayscale) images. In this section, we apply the same strategy to RGB images (three channels). To do this, we adapted the previous code by splitting the image into its three individual channels, applying the blurring and deconvolution operations to each channel separately, and finally stacking them to form the recovered RGB image (see Algorithm 1).

The execution is the same as we did in its respective section, obtaining results like the ones seen in the figure 51.

Algorithm 1 RGB Image Recovery Process

```

1: Input: Aperture image, RGB image, noise level  $\sigma$ , blur size
2:
3: Output: Recovered images using different deconvolution methods
4: // Load and process the data
5: // Process each channel separately
6: for  $c \leftarrow 1$  to channels do
7:    $f1(:, :, c) \leftarrow \text{zDefocused}(\text{image}(:, :, c), k1, \sigma, 0)$ 
8:    $f0\_hat(:, :, c) \leftarrow \text{zDeconvWNR}(f1(:, :, c), k1, C)$ 
9:    $f0\_hat\_wnr(:, :, c) \leftarrow \text{deconvwnr}(f1(:, :, c), k1, \sigma^2)$ 
10:   $f0\_hat\_lucy(:, :, c) \leftarrow \text{deconvlucy}(f1(:, :, c), k1, 10)$   $\triangleright 10$  iterations
11: end for
12: // Display Results
13: Combine channels to form final recovered RGB images
14: Display original, defocused, and recovered images for comparison
  
```



Figure 48: $\sigma = 0.0001$ and 10 blur size



Figure 49: $\sigma = 2$ and 2 blur size



Figure 50: $\sigma = 10$ and 10 blur size

Figure 51: Comparison between both extreme cases where we can put or high noise (middle) or blur size (upper) and we can put both high (bottom) for an RGB image

Those results suggest that processing RGB images is just a generalization of those methods applying them 3 times (one per channel) without improving the results as they are equally the same results.

Additionally, rather than recalculating the heatmaps, we will plot the metrics using a line graph to evaluate how they vary with respect to the blur size and MSE (Fig. 55)

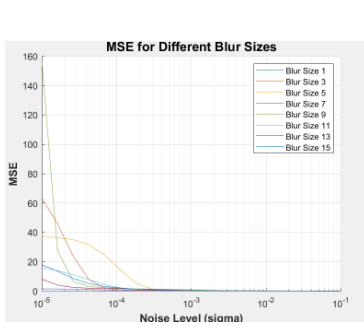


Figure 52: Wiener with prior

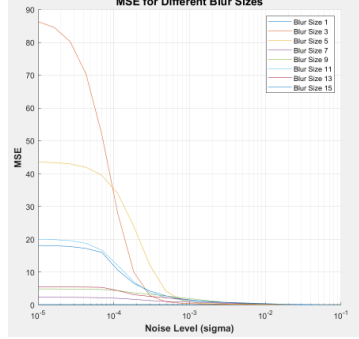


Figure 53: Wiener without prior

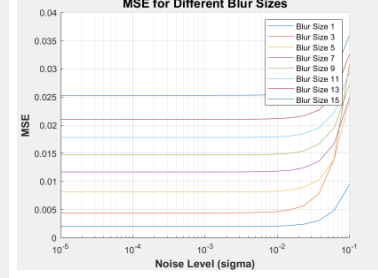


Figure 54: Lucy-Richardson

Figure 55: MSE of the recovered images with respect to the original one with different deconvolution strategies.

From the images, we can see that for the Wiener deconvolution with priors, the error decreases more rapidly compared to the version without priors, as the prior information helps to better recover the image. For Wiener without priors, we observe that as both noise levels and blur size increase, the results tend to approach the original image more closely. On the other hand, Lucy-Richardson remains stable at first as noise increases but eventually starts to degrade as the noise continues to rise, contrasting with the behavior of the Wiener methods.

6 Conclusion

References

- [1] Wikipedia contributors. Psnr, 2025. Accessed: 2025-02-14.
- [2] Wikipedia contributors. Structural similarity index measure, 2025. Accessed: 2025-02-14.